

 [View source on GitHub \(https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L2883-L2939\)](https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L2883-L2939)

Creates a tensor with all elements set to one (1).

```
tf.ones(  
    shape,  
    dtype=tf.dtypes.float32 (https://www.tensorflow.org/api_docs/python/tf/dtypes#float32),  
    name=None,  
    layout=None  
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Validating correctness & numerical equivalence (https://www.tensorflow.org/guide/migrate/validate_correctness) • Use TF1.x models in TF2 workflows (https://www.tensorflow.org/guide/migrate/model_mapping) • Better performance with tf.function (https://www.tensorflow.org/guide/function) • Extension types (https://www.tensorflow.org/guide/extension_type) • Ragged tensors (https://www.tensorflow.org/guide/ragged_tensor)	• Multilevel Modeling Primer in TensorFlow Probability (https://www.tensorflow.org/probability/examples/Multilevel_Modeling_Primer) • Learnable Distributions Zoo (https://www.tensorflow.org/probability/examples/Learnable_Distributions_Zoo) • Probabilistic PCA (https://www.tensorflow.org/probability/examples/Probabilistic_PCA) • Policies (https://www.tensorflow.org/agents/tutorials/3_policies_tutorial) • Bayesian Switchpoint Analysis (https://www.tensorflow.org/probability/examples/Bayesian_Switchpoint_Analysis)

See also [tf.ones_like](https://www.tensorflow.org/api_docs/python/tf/ones_like) (https://www.tensorflow.org/api_docs/python/tf/ones_like), [tf.zeros](https://www.tensorflow.org/api_docs/python/tf/zeros) (https://www.tensorflow.org/api_docs/python/tf/zeros), [tf.fill](https://www.tensorflow.org/api_docs/python/tf/fill) (https://www.tensorflow.org/api_docs/python/tf/fill), [tf.eye](https://www.tensorflow.org/api_docs/python/tf/eye) (https://www.tensorflow.org/api_docs/python/tf/eye).

This operation returns a tensor of type `dtype` with shape `shape` and all elements set to one.

```
>>> tf.ones([3, 4], tf.int32)  
<tf.Tensor: shape=(3, 4), dtype=int32, numpy=  
array([[1, 1, 1, 1],  
       [1, 1, 1, 1],  
       [1, 1, 1, 1]], dtype=int32)>
```

Args	
shape	A list of integers, a tuple of integers, or a 1-D Tensor of type <code>int32</code> .
dtype	Optional DType of an element in the resulting Tensor . Default is tf.float32 (https://www.tensorflow.org/api_docs/python/tf#float32).
name	Optional string. A name for the operation.
layout	Optional, tf.experimental.dtensor.Layout (https://www.tensorflow.org/api_docs/python/tf/experimental/dtensor/Layout). If provided, the result is a DTensor (https://www.tensorflow.org/guide/dtensor_overview) with the provided layout.

Returns
A Tensor with all elements set to one (1).

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